

Umberto Domenico Ciccia

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Experience

Amazon Web Services (AWS), System Dev Engineer

Dublin, Ireland Nov 2025 – present

- Unblocked a strategic customer launch by delivering DNSSEC support for Route 53 in AWS European Sovereign Cloud 3 weeks ahead of schedule, allowing the customer to migrate without modifying their existing infrastructure.
- Maintained reliability of Route 53 services in sovereign regions, achieving 99.999% availability through incident response and RCA-driven improvements.
- Supported infrastructure deployment for AWS European Sovereign Cloud, enabling compliant in-region operations.

NTT Data Italia S.p.A., Cloud Engineer

Cosenza, Italy Jan 2025 – Oct 2025

- Built CI/CD pipelines on GCP to deploy Dataflow pipelines processing 10k+ daily records, improving data availability and release reliability.
- Designed multi-cloud infrastructure (AWS + on-prem) using Terraform and Kubernetes, improving deployment consistency and reducing failure rates by 70%.
- Developed microservices (Quarkus, Crossplane) to automate infrastructure policy management and optimize cloud resource allocation.

Vision One S.r.l., Software Engineer

Recanati, Italy July 2024 – Dec 2024

- Developed scheduling system in .NET/Blazor, improving task allocation efficiency by 80%.
- Refactored legacy codebase, reducing unused code by 60% and improving maintainability.
- Introduced automated testing (xUnit), increasing test coverage to 80%.

Caliò Informatica S.r.l., Software Engineer

Cosenza, Italy Mar 2024 – July 2024

- Developed workflow-building capabilities for AIDA, an OpenAI-powered automation platform, enabling non-technical users to create AI workflows through a visual drag-and-drop interface inspired by n8n.
- Optimized OpenAI API consumption through request redesign and prompt engineering strategies, reducing operational costs by 80% while improving response performance by 60%.

Projects

AWS MLOps github.com/umbertocicciaa/aws-mlops

- Built end-to-end ML pipeline (Terraform, SageMaker) for training, evaluation, and real-time inference.
- Automated model promotion and deployment using Lambda and performance thresholds.

Italian Db github.com/umbertocicciaa/italian-db

- Designed and implemented an educational DBMS from scratch to explore core database internals, including query execution, storage management, and data structure design.

Raspberry Pi 4 Kubernetes Cluster github.com/umbertocicciaa/homelab-utils

- Designed and deployed a 2-node K3s cluster with CI/CD, ingress (Traefik), and observability stack (Prometheus, Grafana).
- Implemented secure networking (DNS, SSH, static IPs) and production-style workload deployment.

Skills

Certifications: AWS Solutions Architect Associate, Certified Kubernetes Administrator (CKA), Google Associate Cloud Engineer, Terraform Associate

Languages: Python, C#, Java, Bash, TypeScript, C, GO

Cloud & DevOps: AWS, GCP, Azure, Kubernetes, Terraform, Docker, CI/CD

Backend & Frontend: .NET, Spring Boot, Quarkus, Django, React, Angular, Blazor

Databases: PostgreSQL, SQLServer, MongoDB, Redis

Education

M.Sc. Università della Calabria, Cyber Security Engineering

B.Sc. Università della Calabria, Computer Engineering

Extra Activities and Personal Achievements

- Won the AWS Jam for NTT Data.
- Participated in the MLSysOps Hackathon, designing Kubernetes-based policies for reliable ML deployments with Prometheus/Grafana observability, autoscaling, and automated rollouts.
- Lead The Future Mentee, a highly selective STEM mentorship program with an acceptance rate below 20%.
- Maintainer of Unical<3Telegram, GitHub platform used by 2000+ DIMES students for exam preparation and resource sharing (unicallovetelegram.github.io/IndiceArgomentiOrale).